Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/16/2023
PI Name:	Driskell, James Ph.D.		
Project Title:	Dyads and Triads at 140 Million Miles: Factors Affecting Interpersonal Relations in Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32169-4817	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	01/06/2021	**End Date:**	01/06/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/06/2026 per A. Beitman/NASA JSC (Ed., 9/30/24). NOTE: End date changed to 01/05/2025 per A. Beitman/NASA JSC (Ed., 11/29/23).		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Driskell, Tripp Ph.D. (Florida Maxima Corp) Burke, Shawn Ph.D. (University of Central Florida)		
Grant/Contract No.:	80NSSC21K0457		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	September 2024 Update: Per NASA's Human Factors and Behavioral Performance Element, a one-year augmentation to this grant added an Antarctic analog and 1 year to the period of performance (POP). The Driskell team is augmenting their NASA research by utilizing the Amundsen-Scott, McMurdo, and Palmer stations as an analog for teams operating in long duration, isolated, confined environments (Ed., 9/30/24). Psychosocial dynamics within the context of spaceflight have been shown to have a large impact on crew dynamics and well-being. As NASA moves towards longer duration exploration missions, the isolation and confinement may foster a blurring of work and social factors. The increased social support provided by crew members along with increased social interdependence has the potential to cause differential levels of interpersonal and/or intimate relationships among subsets of the crew. We propose a multi-faceted program of research to identify how trigger events will disrupt interpersonal relationships and corresponding markers of factors fostering social resilience. We will develop, test, and validate an individual-level and team-level countermeasure to facilitate social resilience in terms of accurate assessment of affect-based team dynamics and mitigate adverse effects of shifts in team dynamics due to interpersonal relationships.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Project deliverables will document: (1) factors associated with the development and breakdown of social resilience at the dyad and team levels across a range of interpersonal relationships, as well as characterization of the risk posed to team functioning and performance, (2) strategies and countermeasures to foster social resilience and mitigate the risk of dysfunctional interpersonal relationships that may threaten team coordination and cohesion, (3) the implementation and evaluation of a set of countermeasures with respect to their effectiveness, acceptability, and feasibility for use, and (4) recommendations and guidelines for the implementation of developed countermeasures within mission scenarios
Task Progress:	Note: Per the Principal Investigator (PI), there are currently seven (7) research tasks where progress is measured in accordance with the project timetable. (Ed., 12/21/2023) Technical progress includes the following tasks: Task 4: Archival Analyses. We have conducted an archival analysis of existing communication data from crews participating in prior NASA analog studies. The purpose of this research is to examine specific measures of interpersonal dynamics which relate to crew interpersonal relations, subgroup formation, and cohesion. Results indicate that these measures allow us to identify stronger, or weaker, relationships which exist within crews, the possible formation of subgroups within the crew, and the impact on crew functioning and performance. This task has been completed and a draft publication is under review. Task 5: Laboratory Study: Drivers/Inhibitors of Subgroup Formation. Sentiment relations between team members may vary according to the strength and type of ties between them. Thus, team member A may develop stronger friendship relations with team member B, relative to others in the group, leading to subgroup formation. These subgroups or cliques, characterized by high rates of voluntary interaction, positive interpersonal sentiments, and normative consensus can lead to feelings of rejection or exclusion among other team members and negative socio-emotional responses such as anger, jealousy, and reduced team cohesion. The purpose of this laboratory study is to examine those factors which promote and inhibit subgroup formation within crews, and to examine alternative countermeasures to minimize subgroup formation. Data collection for this study is currently ongoing. Task 6. Development and Design of the Detect, Integrate, and React (DIR) Countermeasure. The primary product of this task will be a comprehensive and validated countermeasure designed to mitigate adverse effects of disturbances in interpersonal relationships within crews and sustain individual and team well-being in future long-duration missions. To achieve this goal, two sub-tasks have been initiated. Sub-Task 1: The first sub-task was to gain subject-matter expert insight into factors which may support positive interpersonal relations among crewmembers during long-duration spaceflight and factors which may negatively impact crew interpersonal relations, as well as gaining insight into their perceptions of potential countermeasures to support positive interpersonal relations. We have conducted semi-structured interviews with current and retired astronauts in a procedure approved by the NASA Institutional Review Board (IRB) and NASA Human Research Program (HRP) internal boards. To date, interviews have been conducted with 20 prior astronauts, representing over 1,895 days in space. Thematic analyses of these interviews have been conducted, addressing the primary topics of (1) work or role relations, (2) friendship relations, (3) close or intimate relations, and (4) countermeasure development. Sub-Task 2: The second sub-task is the development of the Detect, Integrate, and React (DIR) countermeasure, which is intended to enhance the ability of crewmembers to detect potential breakdowns in interpersonal relations, use this information to reason about interpersonal dynamics, and respond in adaptive ways to promote social resiliency and team cohesion. This task is in progress; the program of instruction has been developed, the training development software selected, and the training modules are being created.
Bibliography Type:	Description: (Last Updated: 01/09/2026)
Papers from Meeting Proceedings	Driskell T, Burke S, Driskell J. "Astronaut perceptions of interpersonal relations in long-duration spaceflight." 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. , Jan-2023
Papers from Meeting Proceedings	Käosaar A, Burke CS. "The Role of Emotional Expressivity in Interpersonal Relationships in Long-Duration Spaceflight. " 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. , Jan-2023
Papers from Meeting Proceedings	Käosaar A, Burke CS, Amon MJ. "Nonlinear Dynamics of Team Affect in Long-Duration Spaceflight." 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. 2023 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 7-9, 2023. , Jan-2023

Papers from Meeting Proceedings	Kaosaar A, Szabó K, Amon MJ, Driskell T, Driskell J, Burke CS. "Do Fluctuation Patterns of Affective States Indicate Differences in Teamwork Outcomes – Utilizing the Potential of Longitudinal Data for Manned Spaceflight." 18th Annual INGRoup Conference, Seattle, WA, July 20-22, 2023. 18th Annual INGRoup Conference, Seattle, WA, July 20-22, 2023. , Jul-2022
Papers from Meeting Proceedings	Käosaar A, Burke CS, Driskell T, Fiore S, Landon LB, Marques-Quinteiro P. "Crew Dynamics Support in Long-Distance Exploration Missions: Where Are We? What Are We Missing? " 18th Annual INGRoup Conference, Seattle, WA, July 20-22, 2023. 18th Annual INGRoup Conference, Seattle, WA, July 20-22, 2023. , Jul-2022